



Meerkat Sentry

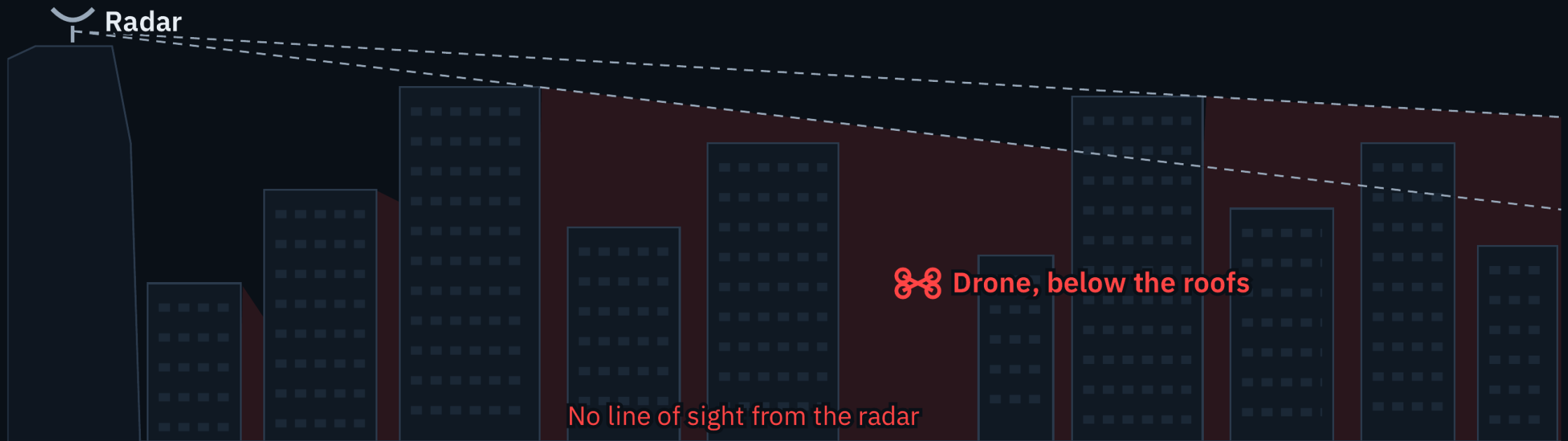
Every roof keeps watch.

TEAM 08 · EDTH TAIPEI 2026 · CHALLENGE 04

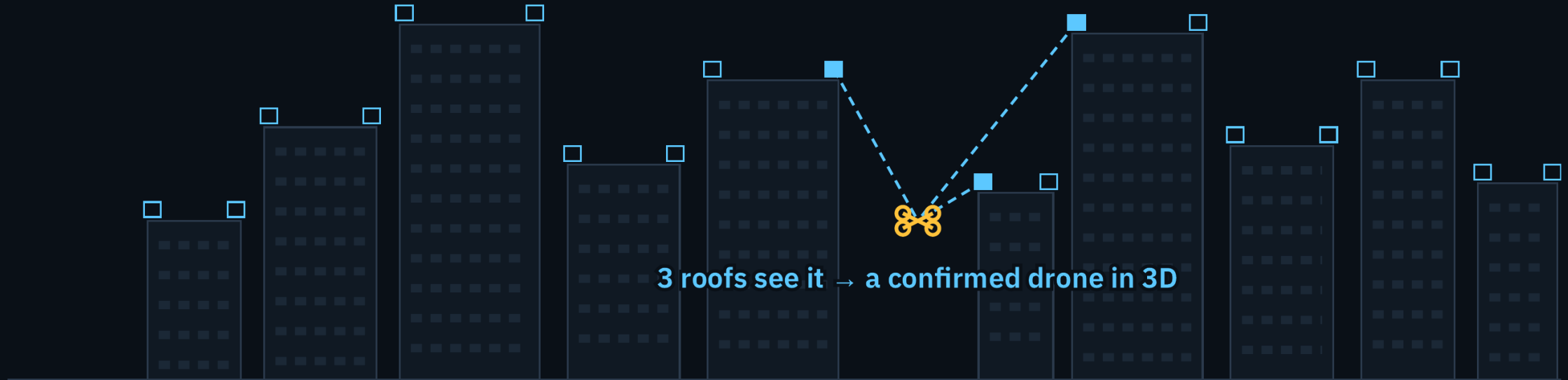
THE PROBLEM

In a city, small drones fly below the roofs, where radar cannot see

Ukraine shows the pattern: cheap drones hit power stations first, then the crews who come to repair them. In Taipei, those sites sit between high-rises.



Cameras on many roofs look along the streets



1 · ROOF BOX

2–6 cheap cameras per roof.
Video stays in the box; only
detections leave it.

2 · CONTROLLER

2 roofs give a 3D position, 3
confirm a drone. Birds, balls and
registered drones are filtered.

3 · OPERATOR

One alert with the reason: speed,
heading, time to target. Accept,
reject or change.

4 · PHONES

People in the drone's path are
told to go inside, through
Taiwan's Public Warning System.

REAL · BUILT AT EDTH, 2-4 OCT

Four parts, all built this weekend

ROOF BOX

Raspberry Pi + camera

Finds the drone in its own image at 15–20 fps.

CONTROLLER

Laptop

Combines the views into one 3D track, 23 ms after the camera.

DASHBOARD

Browser

Rules, alert with the reason, operator decision, log.

WARNING

Phone

Demo: Telegram. Real use: Taiwan's Public Warning System.



OUR LAB · 3 CAMERAS MARK THE SAME DRONE · 20 OF 20 INDOOR FLIGHTS DETECTED

Off-the-shelf parts. No core part made in the PRC.

HOW DETECTION WORKS

Pixel changes reveal moving targets

Each fixed camera compares consecutive frames.

01 PREVIOUS FRAME



$t - 1$

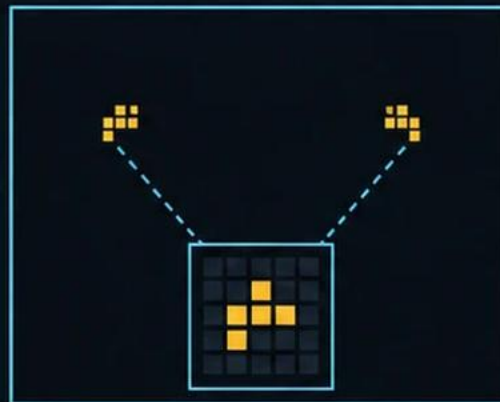
02 CURRENT FRAME



t

Same background, new target position.

03 PIXEL DIFFERENCE



Highlight pixels that changed.

04 MOTION CANDIDATE



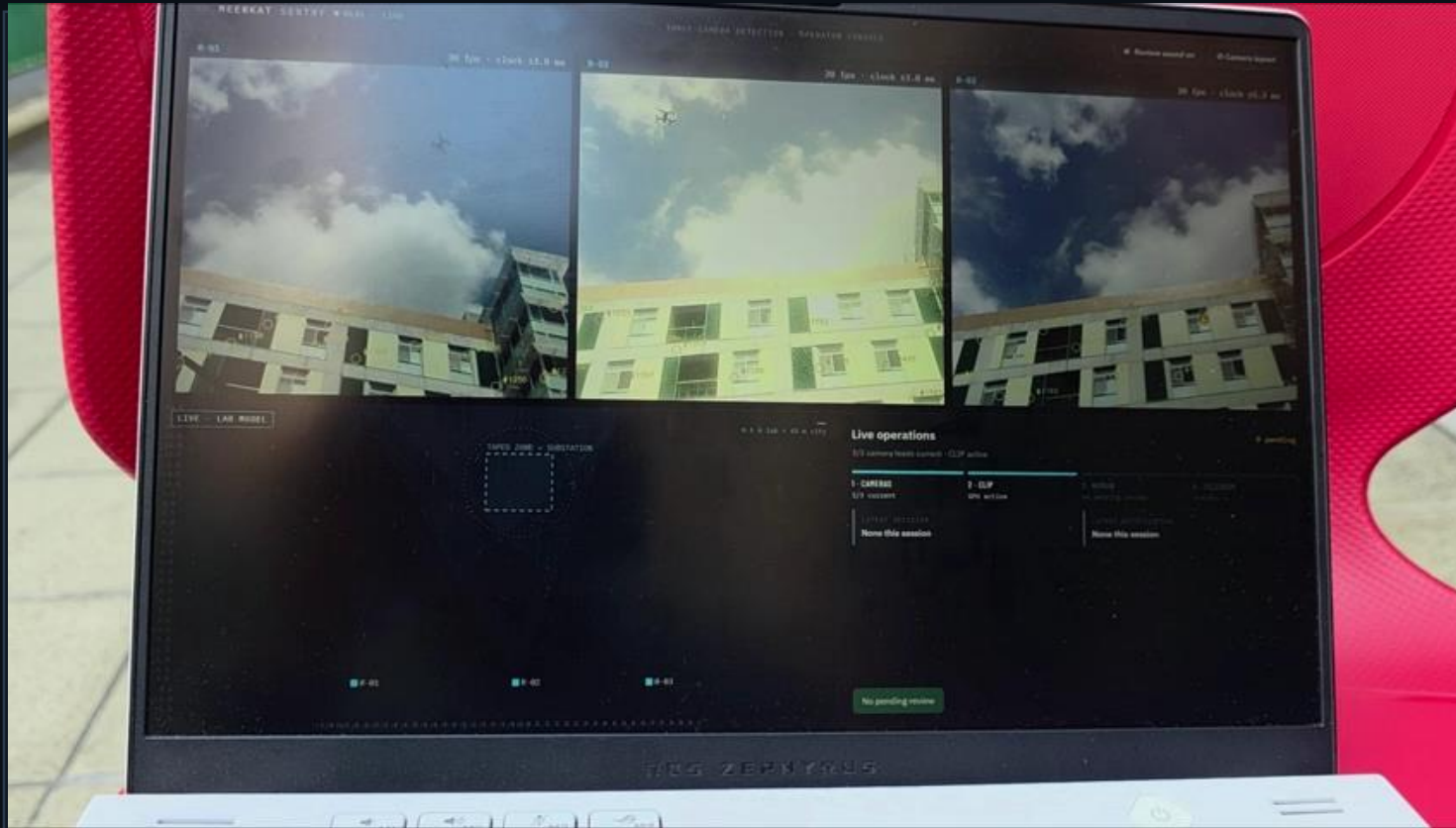
Group changes into a moving candidate.

Pixel change detects motion.

Multiple camera views help confirm the target and estimate its 3D position.

REAL · RECORDED IN OUR LAB

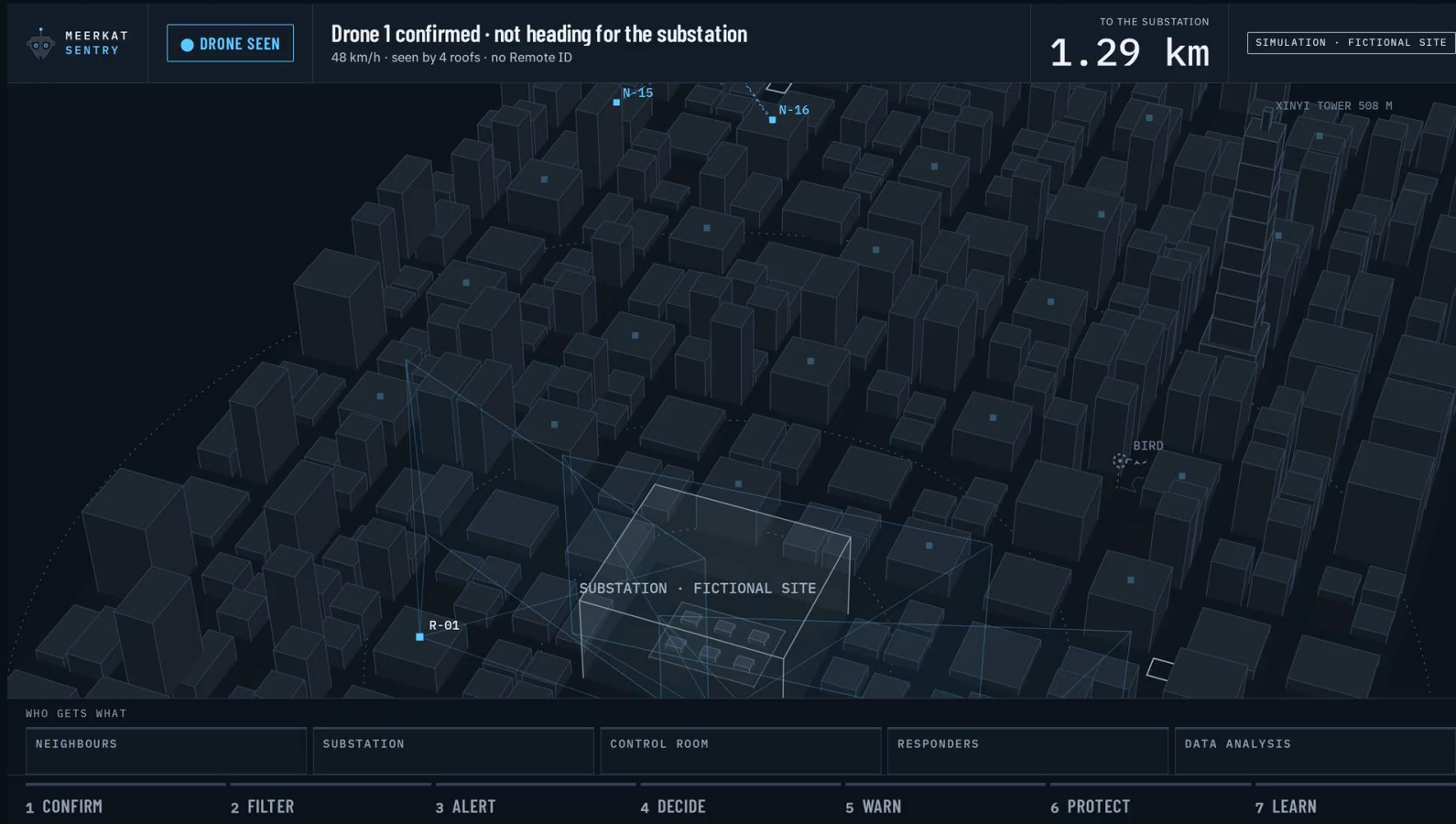
Demo: one drone, from camera to phone



- 1 The drone shows up in the three camera images
- 2 The controller turns them into one track in 3D
- 3 The dashboard raises the alert, with the reason
- 4 The operator approves; the phone gets the warning

SIMULATION · FICTIONAL NEIGHBOURHOOD

The same software, for a city neighbourhood



24 roofs

about 300 m apart

1.4 km

drone confirmed this far from the substation

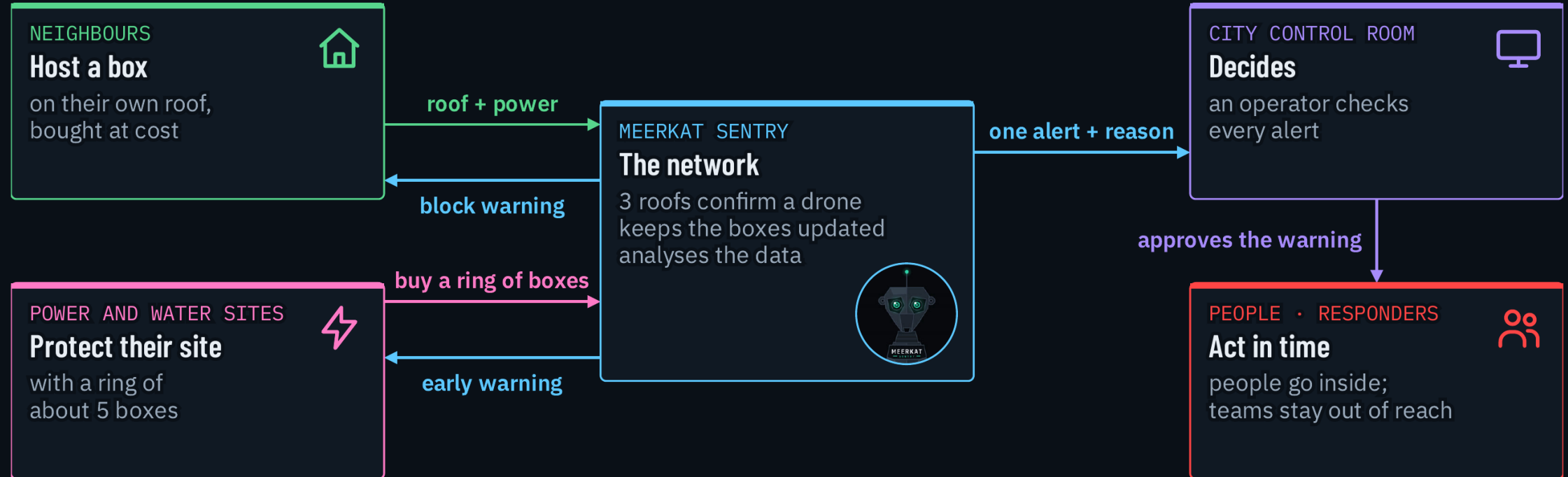
34 s

to get inside if the warning is approved early; 26 s if late

Responders safe

warned of a second drone waiting for them

Neighbours, sites and the city work as one network



Why people join: protecting your own block is part of Taiwan's whole-of-society defence.

Neighbours keep watch over their own city

MAIN CASE · COMMUNITY PROJECT

- 1

On your roof

a small camera box
- 2

Plug it in

one cable for power and network
- 3

Forget it

it updates itself; alerts come to your phone

YOU GET	an alert when a drone is at your window, and a warning for your block in a crisis
IT COSTS	only the box, at cost: \$345 today, ≈ \$100 target at volume
WHY IT WORKS	people defend their own city. Ukraine’s Sky Fortress runs on ~10,000 cheap sensors.

Assumed: \$100 box at volume (one camera chip instead of Pi + AI board), \$8 running cost per box and month. Home alarms in Taiwan: NT\$1,500–2,500 a month. \$2.8 bn: drone detection market (MarketsandMarkets).

OPTION 2 · SUBSCRIPTION

NT\$1,200 / month

≈ \$38 · 12-month minimum



months 1–12: box + running costs 13 →

From month 13 the profit pays for repairs and for processing and analysing the data to improve the network. The data is not sold.

≈ \$36M / year

if all ~78,000 boxes for Taiwan’s cities subscribe. Market: \$2.8 bn by 2030.

Help us put the first boxes on Taipei's roofs

We are looking for partners in the city, at critical sites and on the roofs.
Every new roof makes the network stronger.

Jan Barkemeyer

[linkedin.com/in/janbarkemeyer](https://www.linkedin.com/in/janbarkemeyer)

Robin Fischer

[linkedin.com/in/robin-fischer-104627132](https://www.linkedin.com/in/robin-fischer-104627132)

Edoardo Dominikus

[linkedin.com/in/edoardo-dominikus](https://www.linkedin.com/in/edoardo-dominikus)

Sean Ching

[facebook.com/share/14uKUWWfeEe](https://www.facebook.com/share/14uKUWWfeEe)

Ching Hsu



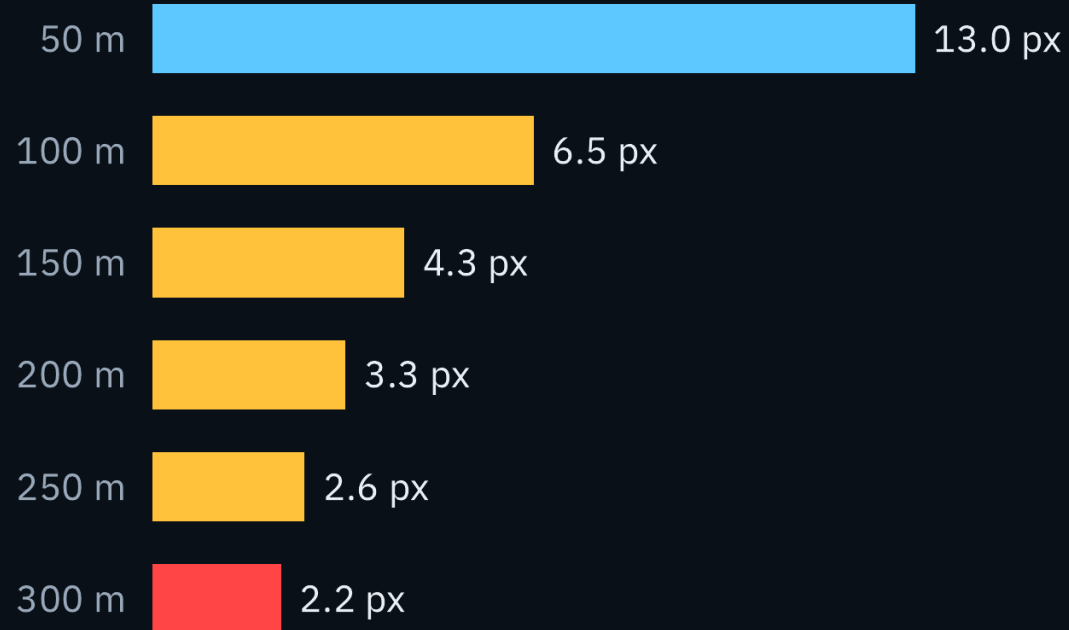
Challenge 04, point by point

CHALLENGE 04 ASKS FOR	WHAT WE BUILT	STATUS
Detect drones in video	3 Pi cameras, controller; 20 of 20 indoor flights	built
Persistent track ID and path	Track IDs and recorded path in the dashboard	built
Alerts: speed, direction, dwell, zone entry	Each alert states its reason in plain words	built
Recommend: track, warn, avoid, allocate	4 recommendations per alert	built
Operator accepts, rejects, modifies	Buttons and a decision log with times	built
Customer, architecture, scaling	Slides 3, 4, 7 and 8	pitch
No weapon control	Warnings and team allocation only	by design

Field problems and our answers

PROBLEM	OUR ANSWER	STATUS
Birds	Erratic course: filtered before the operator	built
Balls, thrown objects	Fall in an arc: filtered	built
Registered civil drones	Remote ID received: filtered, unless heading for the zone	built
Decoys, one faulty camera	At least 3 independent roofs must agree	built
Night, fog	Cheap cameras cannot see. Next: microphones, thermal	roadmap
Sun in the lens	Cameras face away from the sun; sun sector masked	designed
Hacking	Boxes send only small detection messages; video never leaves the box	designed
Internet down	Tiny messages, neighbour mesh, battery	designed
Typhoon	Sealed box, about 10 W, typhoon mount	designed

How far one camera sees a small drone



Pixels across a 35 cm drone with a wide camera (4608 px over 102°): $\approx 650 \div \text{distance in m}$.

One camera alone needs about 8–10 px: about 80 m.
Several roofs that agree can work with 2–3 px: about 200–300 m.

So the boxes sit 250–300 m apart.

Drone size and pixel thresholds are assumptions.

What the operator never sees

3,752

camera detections

4

tracks

2

drones

1

threat

FILTERED	WHY
60 sensor artefacts	Seen by one camera only
1 ball	Falls in an arc
1 bird	Erratic course
1 registered drone	Sends Remote ID
Never filtered	Anything heading for the protected zone

Numbers from our simulated neighbourhood scenario.

Deciding early gives people more time

Operator pre-approves the warning when the drone turns in



Sent the moment the drone is on course to the site

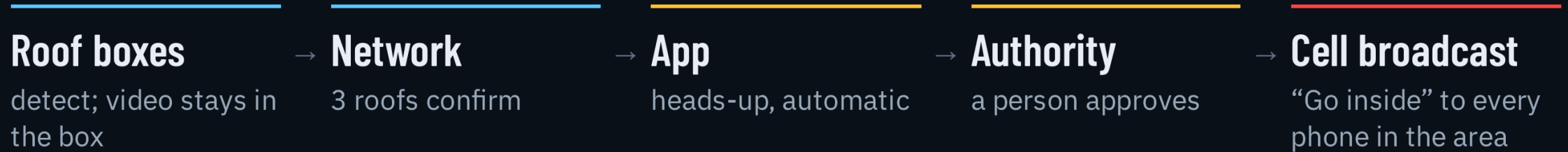
Operator decides at the last moment



8 seconds of thinking are 8 seconds less to hide

Time budget: 30 s to take cover + 10 s delivery + 10 s operator. Simulation: drone confirmed 1.4 km out, final dash from 240 m.

Who decides which warning



Taiwan already has the channel: the Public Warning System reaches every 3G/4G phone in an area within seconds and already sends earthquake alerts. We add a trigger, not a new alert system. Telegram is only our demo.

What is inside the box

PART	PRICE	SOURCE
Raspberry Pi 5, 4 GB	\$96.49	price Jul 2026
AI HAT+ 13 TOPS (Hailo-8L)	\$70	official
2 × Camera Module 3 Wide	\$77	PiShop
PoE HAT for Pi 5	\$19.99	Waveshare
Housing, window, typhoon mount	\$45	estimate
SD card, cables, fan	\$20	estimate
PoE injector	\$15	estimate
Box S total	≈ \$345	

Box M = 2 computers, 4 cameras ≈ \$650 · Box L = 3 computers, 6 cameras ≈ \$950. The PoE HAT is made in China: replace it for government use.

We add to radar, we do not replace it

	PRICE	COVERAGE	EMITS	WARNS PEOPLE
Counter-drone radar	\$30k–300k per unit	Blind behind high-rises	Yes: can be located	No
AnyLeaf Meerkat	\$1,000–1,500 per site	One site	No	No
Meerkat Sentry	≈ \$345 per roof box	A district, grows with every roof	No	Yes: app + cell broadcast

Cameras need no radio signal: they also see fiber-optic and pre-programmed drones, which RF detectors miss.

“Meerkat” is also the name of AnyLeaf’s single-site detector (anyleaf.org/meerkat). Radar prices: ae-defense.com, unmannedairspace.info.

What every incident teaches the network

QUESTION	WHAT WE LEARN	WHAT CHANGES
Where do they come from?	Approach directions, speeds, time of day	Where the next roofs go
Where do they wait?	Drones hovering before the attack	Patrols check those roofs and parks
When does the second drone strike?	Seconds between impact and second strike	Teams stage out of reach
Where are we blind?	Path parts seen by fewer than 3 roofs	Gaps to close first
How fast were we?	Detection → decision → warning → impact	Operator training

Rules: no video, no faces. Stored in Taiwan. Shared only with authorities and site operators, never sold.